

Jacob Zheng

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Mathematics, Honours, Co-op

- **Activities:** WAT.AI, UW Computer Science Club

SKILLS & AWARDS

Languages: TypeScript, Python, JavaScript, C, C++, SQL, Bash

Frameworks / Libraries: Node.js, React, Next.js, TailwindCSS, FastAPI, PyTorch, PostgreSQL

Developer Tools: Git, Linux, Docker, Claude Code

Certifications: Microsoft Certified AZ-900, AI-900; AWS Certified AI Practitioner

Awards: Hackathon Winner (cuHacking)

EXPERIENCE

Software Engineer

Sep. 2025 – Present

Waterloo Reality Labs

Waterloo, ON

- Built a **gesture recognition pipeline** on the Meta XR All-in-One SDK, training a **neural network** on finger curl and flexion sensor data to classify **5+ custom gestures** unsupported by the base SDK.
- Deployed it to UWRL's standalone XR headset with a **per-user calibration tool** adapting thresholds to individual hand sizes, emitting real-time gesture events consumed by downstream XR applications.

AI Research Consultant

Apr. 2026 – Present

MDPharma

- Researched and designed a **Just-in-Time Adaptive Intervention (JITAI)** system for a Gen Z wellness app, fusing **calendar, biometric, and GPS** signals to infer user state and deliver tailored wellness prompts.
- Delivered a working **prototype** and a **technical implementation guide** spanning model selection, training, and deployment, turning the research into an actionable engineering roadmap for the company.

PROJECTS

Multimodal AI Teaching Assistant

Mar. 2026

- Built a **real-time AI teaching assistant** that detects student disengagement from a webcam and adapts its instruction, fusing live **vision, voice, and conversational AI** into a single tutoring loop.
- Implemented disengagement detection from **MediaPipe Face Mesh** landmarks, reaching **90% detection accuracy**, and streamed events over **WebSockets** at **sub-75ms latency** for responsive feedback.
- Delivered spoken tutoring through **ElevenLabs** voice and the **Gemini API**, served from a **FastAPI** backend with a **React / Next.js** frontend.

Multimodal Hands-Free IDE Controller

Jul. 2026

- Built the **Python backend** for a **VS Code extension** enabling contact-free coding from a webcam, splitting control across **gaze** (eye tracking), **gestures** (touchless control), and **voice**, avoiding the false-activation problem of gaze-only input.
- Implemented gaze estimation from **MediaPipe** iris landmarks in a head-normalized face frame, fit with **SVD-projected ridge regression** to calibrate per user from **under 50 samples** in **under a minute**.
- Designed a **hold-gated gesture state machine** assigning left and right hands distinct navigator and editor roles, and exposed it to the **TypeScript** client over a token-authenticated loopback with a **JSON schema**.

Photo Authenticity Verification App

Jul. 2026

- Built a **mobile camera app** that cryptographically proves photo authenticity at the moment of capture, addressing the growing problem of AI-generated and manipulated images.
- Implemented **Ed25519 signing** on-device and anchored signed photo hashes to the **Solana blockchain**, with verification metadata stored in **MongoDB Atlas** for fast, scalable cross-device lookups.
- Designed a **three-tier verification system** (green, amber, grey) combining **SHA-256** exact hashing with **perceptual hashing**, flagging images with more than a **6-byte hash difference** as AI-generated or manipulated, built with **TypeScript / React Native** and a **Rust on-chain program**.